Shunkei Kakimoto

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Education

University of Minnesota — St. Paul, MN
Ph.D., Applied Economics — *Present*

University of Nebraska-Lincoln — Lincoln, NE
M.S., Agricultural Economics — *2022*

Hokkaido University — Hokkaido, Japan
B.S., Agricultural Economics — *2019*

Research Area

Environmental and natural resource economics; agricultural and environmental policy; economics of technology adoption and diffusion; production economics.

Peer-Reviewed Publications

Mieno, T., Foster, T., **Kakimoto, S.**, and Brozović, N. 2024. "Aquifer depletion exacerbates agricultural drought losses in the US High Plains". *Nature Water*, 1-11.

Kakimoto, S., Mieno, T., Tanaka, T. S., and Bullock, D. S. 2022. "Causal forest approach for site-specific input management via on-farm precision experimentation". *Computers and Electronics in Agriculture*, 199, 107164.

Outreach and Extension Publications

Jayasekera, D., Mieno, T., and **Kakimoto, S**. 2026. "The Effectiveness of Wellhead Protection Areas on Groundwater Quality: Evidence from Nebraska." Department of Agricultural Economics, University of Nebraska–Lincoln. April 1.

Work in Progress

Jayasekera, D. H., Mieno, T., and **Kakimoto, S.** "Groundwater quality implications of wellhead protection areas: Evidence from Nebraska"

Gallagher, N. and **Kakimoto, S.** "The Ag-Response Model: For Analyzing Nitrogen Use and Best Management Practice Adoption by Minnesota Crop Farmers" (Submitted)

Teaching Experience

Instructor, University of Minnesota — Summer 2024
Introduction to R Statistical Analysis Software — Course website

Teaching Assistant, University of Minnesota — Fall 2023 – Fall 2024
APEC 1101 – Principles of Microeconomics, Undergraduate
APEC 8211 – Econometric Analysis I, Ph.D.
APEC 8212 – Econometric Analysis II, Ph.D.
APEC 8003 – Applied Microeconomic Analysis of Game Theory and Information, Ph.D.
APEC 8004 – Applied Microeconomic Analysis of Social Choice and Welfare, Ph.D.

Teaching Assistant, University of Nebraska–Lincoln — Fall 2020, Spring 2022
AECN 896-003 – Applied Econometrics, M.S.

Research Experience

Graduate Research Assistant — 2025 – 2026
University of Minnesota

- Evaluated the cost-effectiveness of a pay-for-performance cover crop program in Olmsted County, Minnesota.
- Developed an agricultural production model to evaluate policies designed to promote the adoption of best management practices in Minnesota.

Graduate Research Assistant 2021 – 2022

Daugherty Water for Food Global Institute, University of Nebraska

- Examined the heterogeneous impacts of groundwater allocation limits on groundwater use in Nebraska.
- Examined the impacts of drought and agricultural policies on agricultural land values.

Graduate Research Assistant 2019 – 2021

University of Nebraska–Lincoln

- Applied causal machine learning methods to site-specific input management.

Selected Presentations

2026

"Cost-Effectiveness of a Pay-for-Performance Cover-Crop Program: A Case Study of the Olmsted County Groundwater Protection and Soil Health Program in Minnesota." Minnesota Water Resources Conference, St. Paul, MN. October 2026

"Heterogeneous Dynamic Water-Use Responses to Irrigation Technology Adoption" AAEA Annual Meeting, Kansas City, MO, July 2026

2025

"Size and the Nature of Measurement Error in Gridded Weather Datasets and Its Consequential Estimation Bias in Regression Models: An Application to PRISM Datasets for the U.S. Midwest." AAEA Annual Meeting, Denver, CO, July 2025; AERE Conference, Santa Ana Pueblo, NM, May 2025.

"Yield Response to Nitrogen in Agricultural Production Models." AAEA Annual Meeting, Denver, CO, July 2025.

2021

"Machine Learning Methods for Site-Specific Input Management." Conference on Farmer-Centric On-Farm Experimentation, October 2021.

Seminar & Workshop Presentations

"Reproducibility Practice in Economics." Applied Economics Student Workshop Series, University of Minnesota, April 2026. Workshop materials

"Introduction to Making Presentation Slides with Xaringan." Applied Economics Student Workshop Series, University of Minnesota, April 2024.

Professional Service

Graduate Student Representative, Faculty Search Committee, University of Minnesota Spring 2026
Discussant, Applied Economics Student Seminar, University of Minnesota Spring and Fall 2024

Awards & Fellowships

Watershed Innovations Scholars Program, University of Minnesota Water Resources Center 2026 – 2027
MNDrive Global Food Fellowship, University of Minnesota Fall 2026 – Spring 2027
CFANS Graduate Assistant Teaching Award, University of Minnesota 2025
Teaching Assistant Award for 5000/8000-Level Classes, University of Minnesota 2023 – 2024
Summer Scholarship, University of Minnesota Summer 2023
Graduate Fellowship, University of Minnesota Spring 2023
Mary A. and Robert B. Litterman Fellowship, University of Minnesota Fall 2022
Tomek Outstanding M.S. Student, University of Nebraska–Lincoln Spring 2022

Additional Skills

Programming: Proficient in R, including spatial data processing; working knowledge of Julia, Python, and Stata.

Certification: Minnesota Irrigator Program, Certificate of Completion (2025).

Languages: Japanese (native); English (fluent).